Presentation

Clemens (Host): [00:35] Hey, Yaroslav.

Yaroslav: [00:37] Uh, hello everyone. Yeah, I'm good.

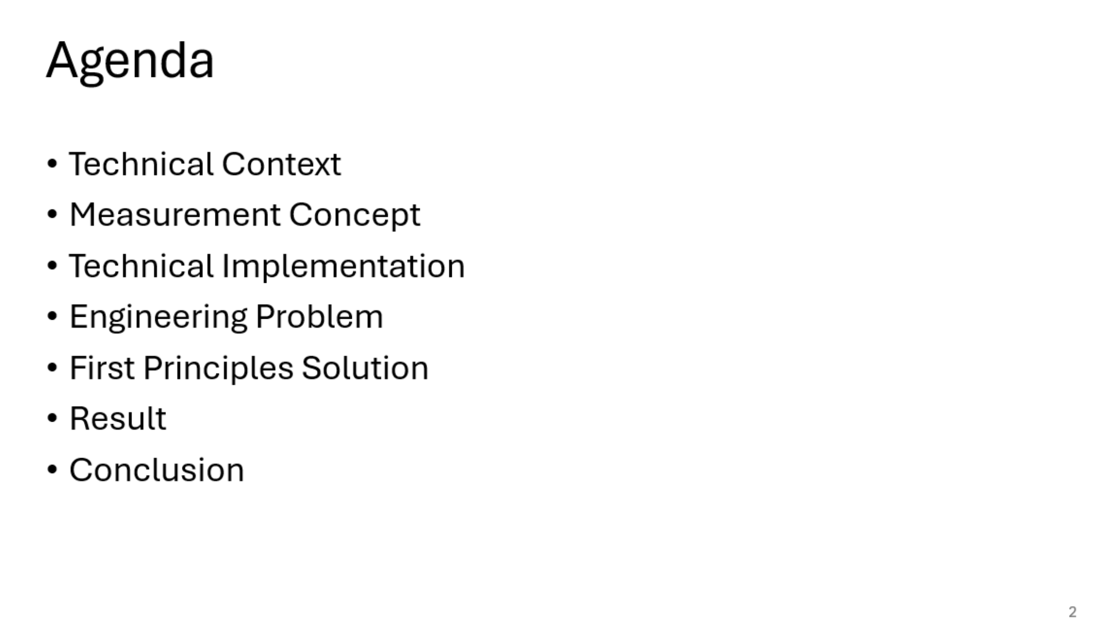
Clemens: [00:43] Yeah.

Yaroslav: [00:44] So, should I share my screen and start?

Clemens: [00:47] Yeah, you can share your screen. Um, we... we're all here like Battery Structures, um, team. Um, I don't think we need to introduce ourselves right now. We already three minutes over, and you will have 1-on-1s with us, um, afterwards anyways. So I would say let's... let's kick it off. Please start. Thanks.

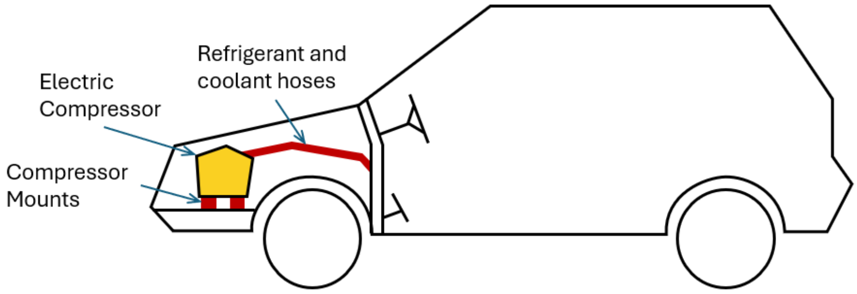
Yaroslav: [01:07] Fine. I'll share my screen. So... Yes. Short introduction about myself. Um, my name is Yaroslav Semenov. I am from Ukraine. I finished high school in Ukraine and moved to Germany to pursue my engineering studies here. I did Bachelor's in Mechanical Engineering, uh, and did an internship at Mahle. I also wrote my Master's... uh, Bachelor's thesis at Mahle. And then I moved to Munich and did my Master's in Automotive Engineering. And throughout my whole, uh, Master's study I worked as a working student at BMW. And also I wrote my Master's thesis at BMW. But today I want to present a project from my times as a working student in the NVH department. And which is... wait a second... so... which is the development of a high-frequency vibration test rig for elastomeric components.

[02:16] Agenda is as follows. So. Uh, first I'll start with technical context, then our measurement concept, then technical implementation, engineering problem, my first principles solution, and finally the result and conclusion.



[02:35] First of all, why do we need a dedicated high-frequency test rig? Our, um, goal was to improve the acoustic comfort in the cabin. Uh, and electric HVAC compressor generally generates a lot of vibrations, which, uh, are transferred through rubber mounts in chassis and through the hoses into the cabin.

Vibration Transfer



Electric compressors generate high-frequency vibrations transmitted to the cabin via mounts and fluid hoses. Accurately characterizing these transmission paths is essential for optimizing NVH and passenger comfort.

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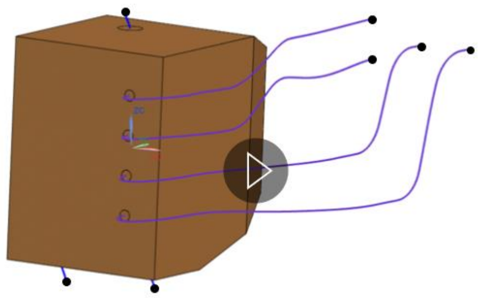
And to solve this, my supervisors developed a simulation workflow which helped us to quantify how much vibrational forces are transmitted into the chassis through the rubber mounts and how much reaches the cabin through the hoses.

Simulation Demo



Simulation workflow quantifies vibration transfer paths through mounts and hoses, enabling data-driven analysis of NVH trade-offs during HVAC system development.

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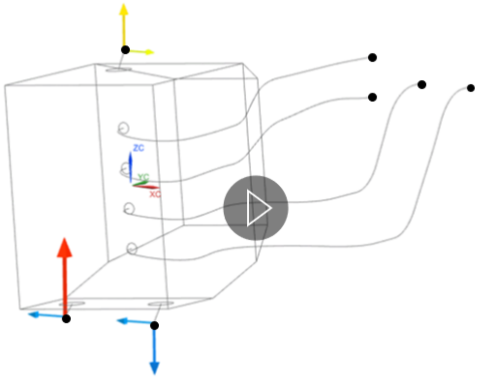
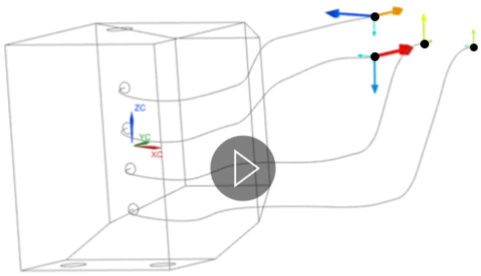
Simulation Demo



Simulation workflow quantifies vibration transfer paths through mounts and hoses, enabling data-driven analysis of NVH trade-offs during HVAC system development.

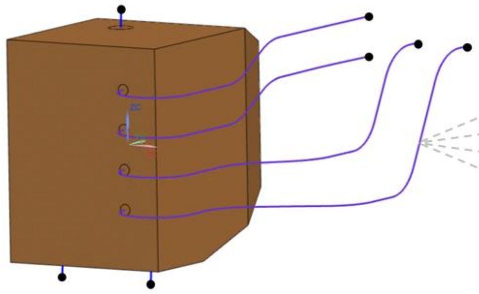
Simulation Demo



Simulation workflow quantifies vibration transfer paths through mounts and hoses, enabling data-driven analysis of NVH trade-offs during HVAC system development.

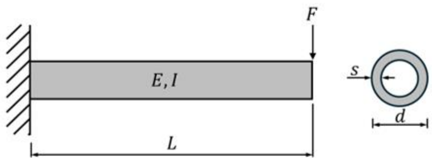
[03:12] However, to make this simulation, uh, predictable, we need, uh, reliable inputs. And by that I mean mechanical properties of the rubber mounts and the hoses. Specifically, we need the axial rigidity, bending rigidity, shear rigidity, and torsional rigidity of the components.

Input Stiffness Parameters



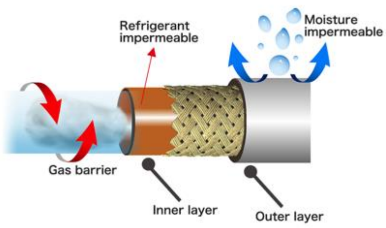
Axial Rigidity (EA)
Bending Rigidity (EI)
Shear Rigidity (GA)
Torsional Rigidity (GI_t)

To enable the simulation workflow, the mechanical behavior of the hoses and compressor mounts must be characterized by four fundamental parameters



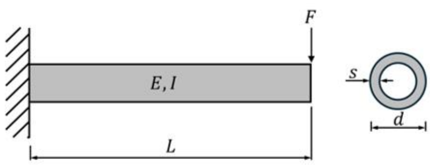
[03:33] You might ask, um: "Yaroslav, why don't... why do we need four different rigidities? Why can't we just take the Young's modulus from the datasheet and, uh, calculate everything, uh, we need based on the geometry of the hoses?" And the answer is yes, that would actually work, but only for the isotropic materials. But since our hoses are, um, composite materials — they have plastic inner layer, they have reinforcement layers and lots of rubber — uh, they just behave differently. If you take the hose, uh, it feels like a, um, wobbly tube. But if you try to twist it, it becames [sic] surprisingly stiff. And that's why we really need the experimental empirical data for the simulation in four different directions.

Input Stiffness Parameters



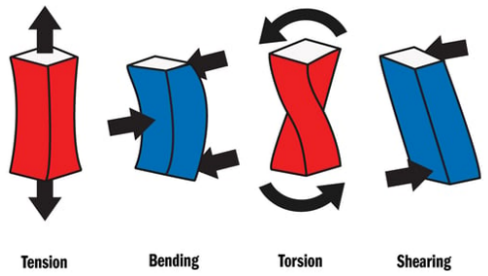
Axial Rigidity ($E_{tension} \cdot A$)
Bending Rigidity ($E_{bending} \cdot I$)
Shear Rigidity ($G_{shear} \cdot A$)
Torsional Rigidity ($G_{torsion} \cdot I_t$)

Complexity of the anisotropic sandwich-structure makes analytical stiffness prediction unreliable, requiring a dedicated experimental approach.



[04:21] You might say: "Fine. Even if we need four different, uh, directions, it should not be that hard, uh, to measure them. Just take the force, apply, um, in the... right... uh... into the desired direction, measure the displacement and there you have it: the stiffness."

Input Stiffness Parameters



Axial Rigidity ($E_{tension} \cdot A$)
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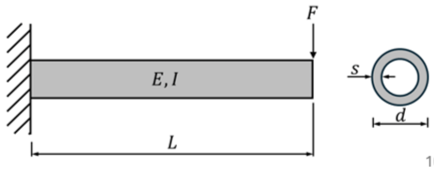
Conventional static measurements offer a straightforward characterization but fail to account the viscoelastic behavior of elastomers.



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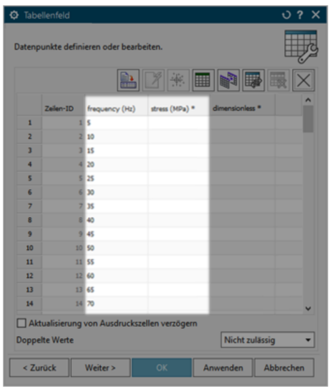
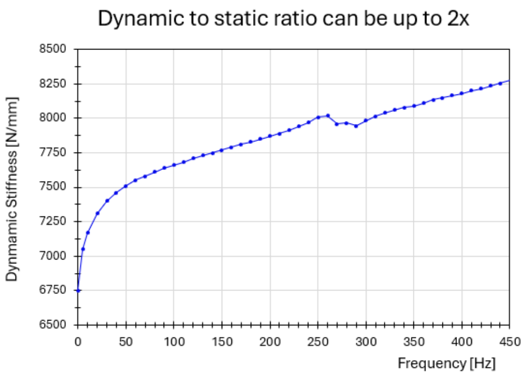
And yes, it would work, uh, but only for materials like metals. But elastomers, unlike metals, they have really annoying feature in terms of NVH, which is viscoelasticity. And that means that if we take, uh, a look at their dynamic stiffness, it is not constant for all conditions, but it is frequency dependent. And, uh, generally the higher the excitation force frequency, the stiffer the sample behaves. Uh, and this dynamic to static ratio at... elas... for elastomers can reach up to 200%. So the difference between dynamic and static, uh, ratio. So to get an accurate simulation, we need as an input not a constant word [sic], but, uh, like throughout... across frequency range, uh, data.

Dynamic Stiffness of Elastomers





Elastomer stiffness is non-linear and frequency-dependent.

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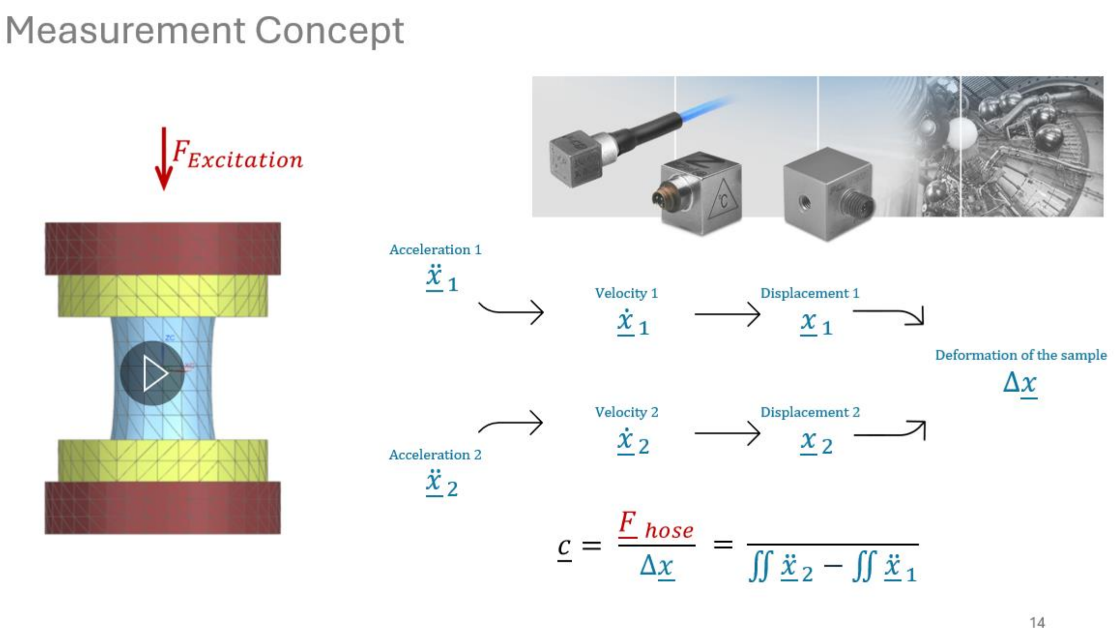
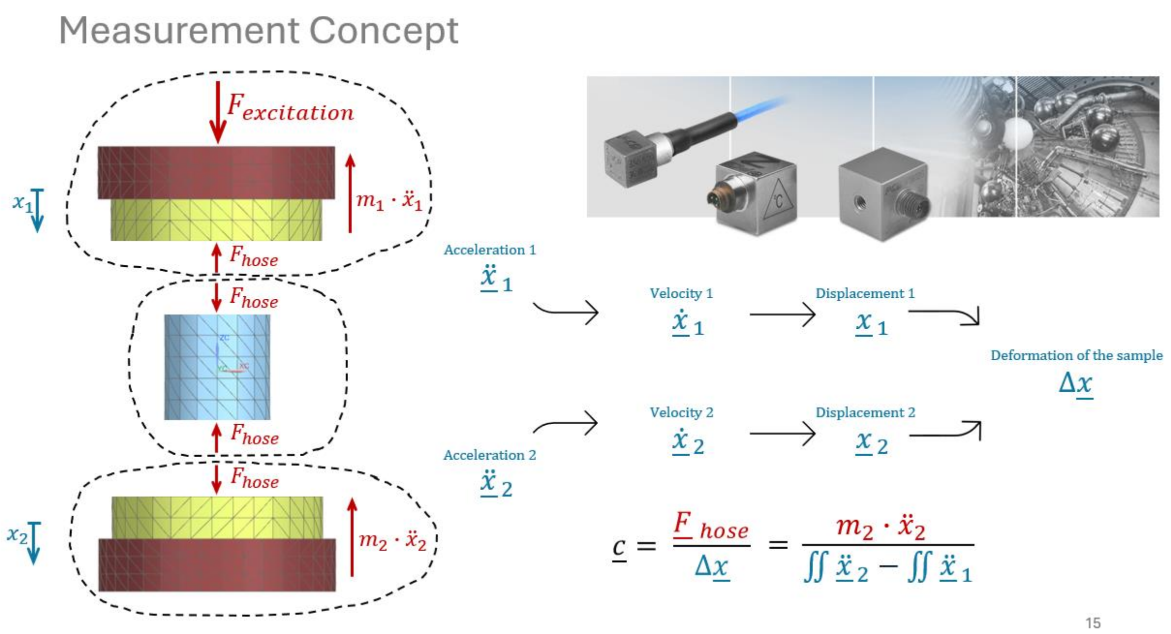
[05:28] Okay. So how do we measure that dynamic stiffness? My supervisors created a new measurement concept, uh, which looks as follows. We have two metal parts with hose between them, uh, and attached to... them... shaker. Uh, so one side is attached to the shaker and other is free floating. And then we attach the acceleration sensors to both parts. And by integrating the, um, acceleration signal numerically over time, we get the velocity of these both parts. And by repeating this step, we can get the displacement of those parts. And by subtracting them, we get the deformation of the sample only by using the accelerometer, uh... sensors.



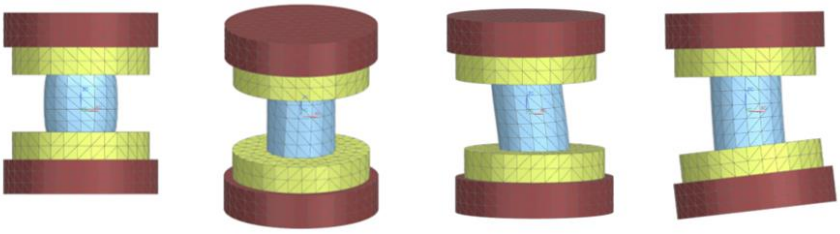
[06:17] Now we need the force, uh... deformation. And to calculate it, we divide our system... we look at it using free body diagram where we have the external forces in form of the shaker, we have the reactional forces between the system, and we also the... have the inertial forces of these two mass bricks. And we take a look... if we take a look at bottom part, we see that here only two forces are acting. That's means if we want to know F hose, we just multiply the... the mass of this brick with its measured acceleration. And here we have the dynamic stiffness measured only using the acceleration sensors.



[07:00] And remember, we have to cover that for four different, um, directions. And that's the beauty of this method because, uh, it eliminates the need of complex and expensive multi-axis load cells. Um, because we can measure with this method, uh, trans... translational, like force and momentum for all types of stiffnesses only using the vibration sensors.

Measurement Concept

Deriving forces and moments from mass properties and acceleration enables multi-directional characterization without the need for complex multi-axis load cells.



$$\underline{c} = \frac{\underline{F}_{hose}}{\Delta \underline{x}} = \frac{m_2 \cdot \ddot{x}_2}{\iint \underline{\ddot{x}}_2 - \iint \underline{\ddot{x}}_1}$$

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[07:27] So let's quickly, uh, wrap up the requirements. It should have four degrees of freedom. It should cover full frequency range, uh, which is 1,000 Hertz in our case. And there was also third important, uh, requirement. And it was operational frequency. Because if I have as an engineer, uh, any question during the HVAC development regarding the material properties, I don't want to spend time on different simulation guesswork. I just want to go downstairs in the lab and quickly measure everything I need without spending days or weeks on, uh, setup. Uh, so...

Requirements Wrap Up

1. Four Degrees of Freedom
2. Full Frequency Range Coverage (up to 1000Hz)

+

3. Operational Efficiency:
   - Fast Setup (< 20min)
   - Empirical Data instead of Simulation Guesswork
   - Testing different sizes, materials, suppliers, temperatures, amplitudes

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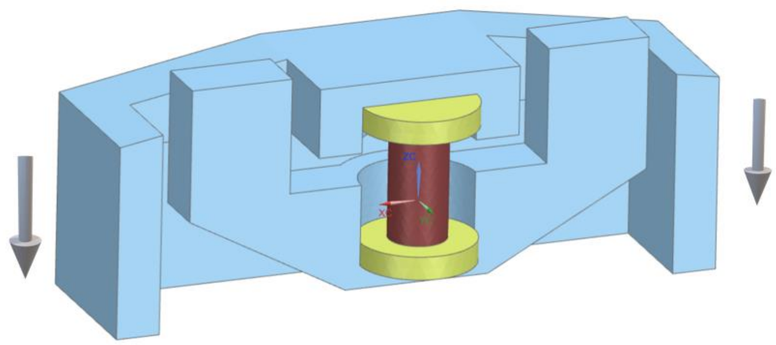
[08:08] And we found a solution how to, uh, do it. At least we thought we found, uh, the solution. And our first, um, iteration... our first concept, uh, looked, um, uh, like that. So how do we build this? Uh, this design was, uh... it is not actual design, it is my simplification for this presentation, but, uh, initial design was developed by my supervisors and then finalized by me. And the idea is as follows. So we have two, uh, parts, uh, active and passive part, and a hose between them. And if we attach two shakers to the outer active part, then when the shakers move in the same direction, we can excite the tension/compression dynamic load on the... our sample.

First Generation



Test rig architecture: the hose is mounted between an outer active part and an inner passive part.

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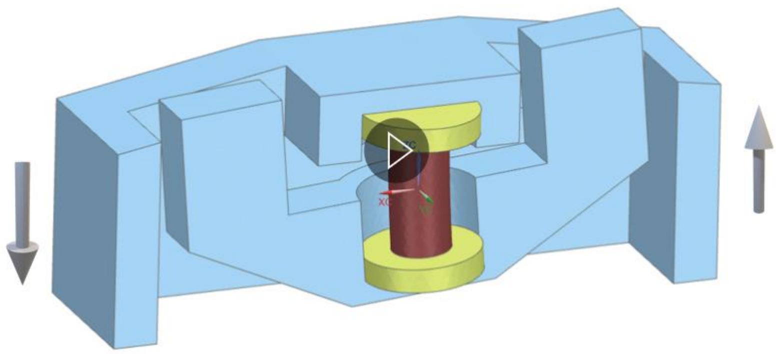
If we simply flip the direction of one shaker, we can excite pure, uh, bending, uh, load on the sample.

First Generation



Four-mode excitation: shaker phase shift and 90° test rig rotation enable rapid Tension, Bending, Torsion, and Shear characterization.

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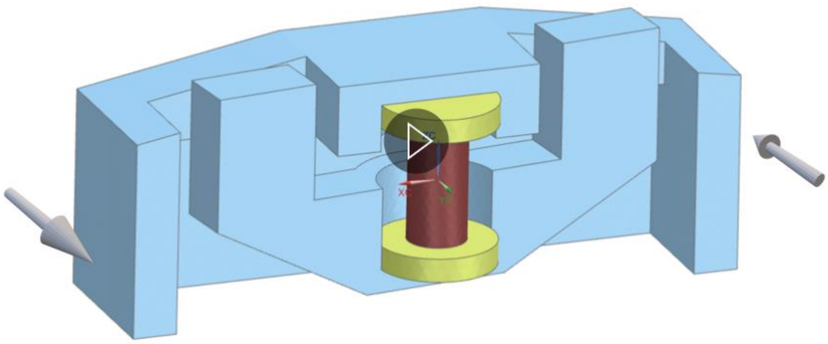
If we rotate this test rig 90 degrees, uh, relatively to the, uh, shakers, uh, we can excite pure torsion load.

First Generation



Four-mode excitation: shaker phase shift and 90° test rig rotation enable rapid Tension, Bending, Torsion, and Shear characterization.

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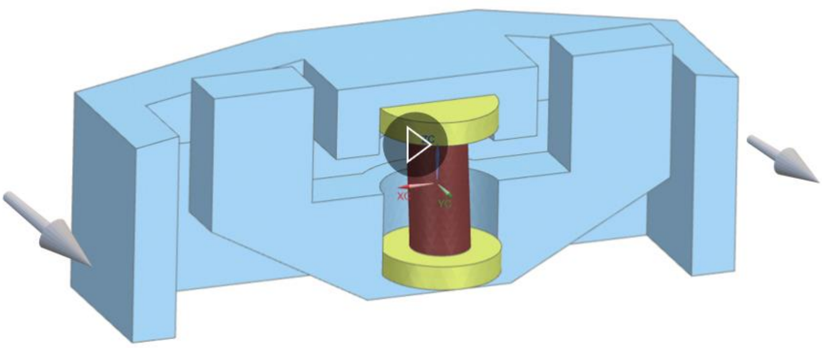
And by flipping the, uh, shaker one more time... the face of the shaker in this new, uh, orientation, we can measure the shear, uh, stiffness.

First Generation



Symmetrical design aligns centers of gravity with the hose axis, eliminating cross-coupling to ensure pure-mode characterization across all load cases.

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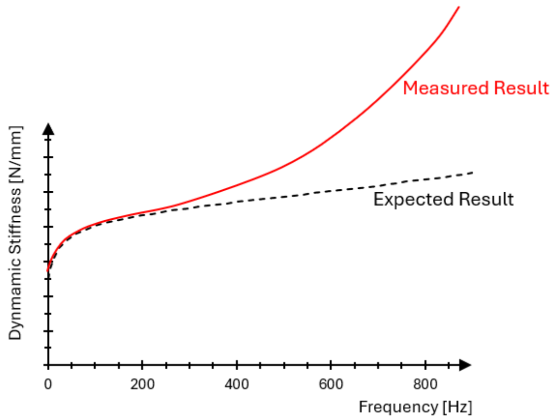
[09:25] So the beauty of this concept was in its symmetry. We can seamlessly transition between different cases. And also, because the center of gravity of the outer part is in the center of the hose, as well as the COG of the passive part in the center of the hose, uh, we can eliminate the, uh, unwanted cross-coupling and induce pure, um, tension, bending, torsion, whatever.

[09:52] Uh, so yes. Um. Symmetrical design. Um. Now, after I covered this, um, context, uh, I can finally explain the engineering problem. So, did it work or did we miss something? I tested this test rig in the lab and the answer is... short answer is yes, it did work, but with a significant limitation. Uh, first of all, um... so if we look to the dy... dynamic stiffness over frequency, we see that it matches our expectations at lower frequencies, but at higher frequencies it just skyrocketed exponentially. Uh, and that’s just not how the elastomers behave in the reality. So something was... was wrong with our measurement. And it was critical because if we feed this artificially high, uh, results into the simulation, we get garbage in, garbage out. And it will force us to over-engineer mounts and refrigerant hoses geometry, uh, solving the "ghost problem" which actually doesn't exist in the vehicle.

Conflict of Theory and Reality



Abnormally high stiffness growth indicates a measurement artifact and leads to “Garbage In, Garbage Out” simulations, resulting in costly over-engineering for non-existent NVH issues.

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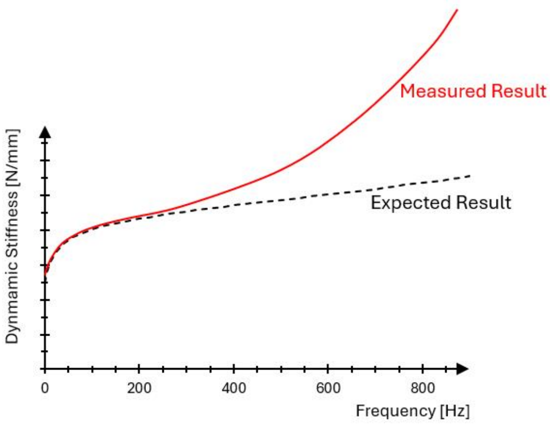
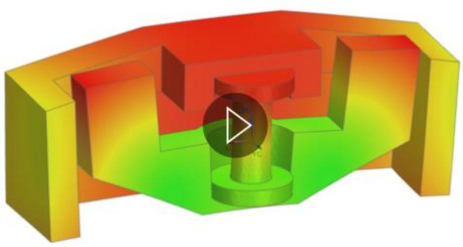
[11:00] So I decided to investigate this problem. And I came to the conclusion that the root cause of that was the system resonance. Uh, like in theory, the test rig should behave as a rigid body. And it actually does at lower frequencies. But the higher we get, the, um... ne... the closer, um, the structural members approaching their natural frequency, uh, deforming and interfering with the sample measurement.

Conflict of Theory and Reality





Root cause: High-frequency system resonances violate the rigid-body assumption, causing fixture deformation that distorts the component’s measured stiffness.

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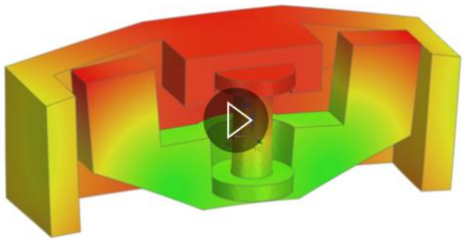
[11:31] Uh, so it was actually good proof of concept, but there was definitely room for improvement. And our next goal was to shift the rig natural frequency far beyond our operational range. Um, ideally five times higher or even ten times higher.

Conflict of Theory and Reality

Goal:

Shift the rig natural frequency far beyond operational range



Root cause: High-frequency system resonances violate the rigid-body assumption, causing fixture deformation that distorts the component's measured stiffness.

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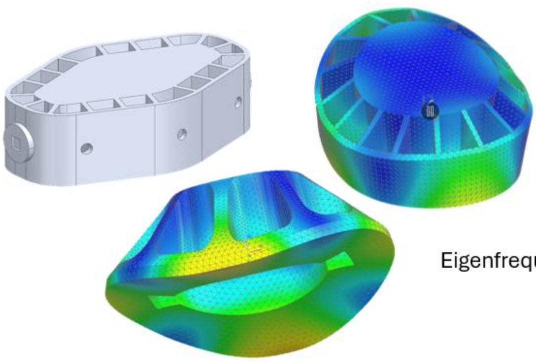
Um. And my supervisor, uh, he tried to solve this problem by adding the reinforcement rift... rib... ribs to the existing structure, um, and to prevent it from deforming. And at some point he came to me and said: "Yaroslav, here your task is to finalize this design and send it to manufacturing."

Battle of ideas



Eigenfrequency = 3 300 Hz

Incremental reinforcement via ribs treats symptoms but fails to sufficiently raise natural frequencies, necessitating a First Principles redesign of the core architecture.

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[12:13] Uh, I took these models, uh, I tested them in the simulation, and somehow I wasn't really convinced. Uh, the natural frequency he achieved was only three times higher than our operational range. And on my opinion, it was just not enough to, uh, eliminate the resonances.

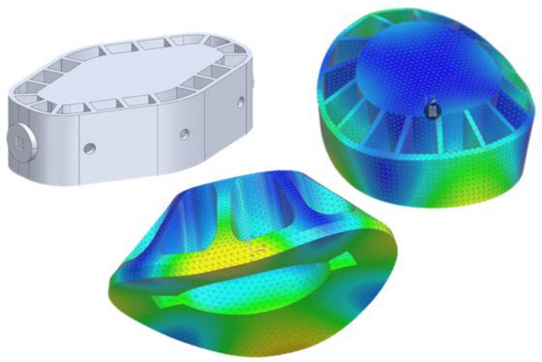
Um, it was... I f... had a feeling that he was trying to treat the sym... symptoms instead of solving the problem. And I believed there was... should be a better solution.

[12:44] So I said to him, uh: "Give me a few days. I want to test few ideas and, uh, create my own design." And at this point he said... uh... "Yaroslav..." I don’t want to sound arrogantly, uh, but I’ve spent on this design, uh, over 40 hours. I tried many iterations and believe me, it is the best solution possible. Uh, and if you try... try it from scratch, uh, then you will just waste your time and you will end up with the solution... with the same result. So just finalize it and send it to production.

[13:20] Um. But for me, it sounded more like a challenge. I was really excited about this problem. And I decided to invest few days of my own time and, uh, I came up with the following ideas. So, if we want to push up the natural frequency, which is the, uh, square root of the stiffness divided by the mass, then it means we want higher stiffness and lower mass. And, uh, while, um, the ribs reinforcement are actually effective in other, uh, engineering problems, but here, uh, they... yes, they increased stiffness... increasing area moment of inertia... uh, but here they were applied only locally on the perimeter and were pointing not always in the desired direction. And in the center we had large opened void which weakened our system. So while the stiffness was higher, um, the second problem was that they increased the part size and, uh, adding more mass. And this additional mass, uh, eliminates the advantages through the system, which results in not very high gain.

[14:32] So I decided to focus entirely on reducing the size of the part and closing the topology. In theory, that should give the double impact on... impact on the natural frequency because first of all, it has the influence on the stiffness in the power of three. And second is that by reducing the size, I drastically reduced the mass which had to, uh, result in a huge gain.

Battle of ideas



$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$\uparrow k \propto \frac{3EI}{L^3}$$

$$\downarrow m$$

Strategy:	Adding ribs	Strategy:	Reduce Size (L) + Close Topology
Result:	$\omega_0 = \sqrt{\frac{k\uparrow\uparrow}{m\uparrow}}$ ≈ Small Gain	Result:	$\omega_0 = \sqrt{\frac{k\uparrow\uparrow\uparrow}{m\downarrow}}$ → Huge Gain

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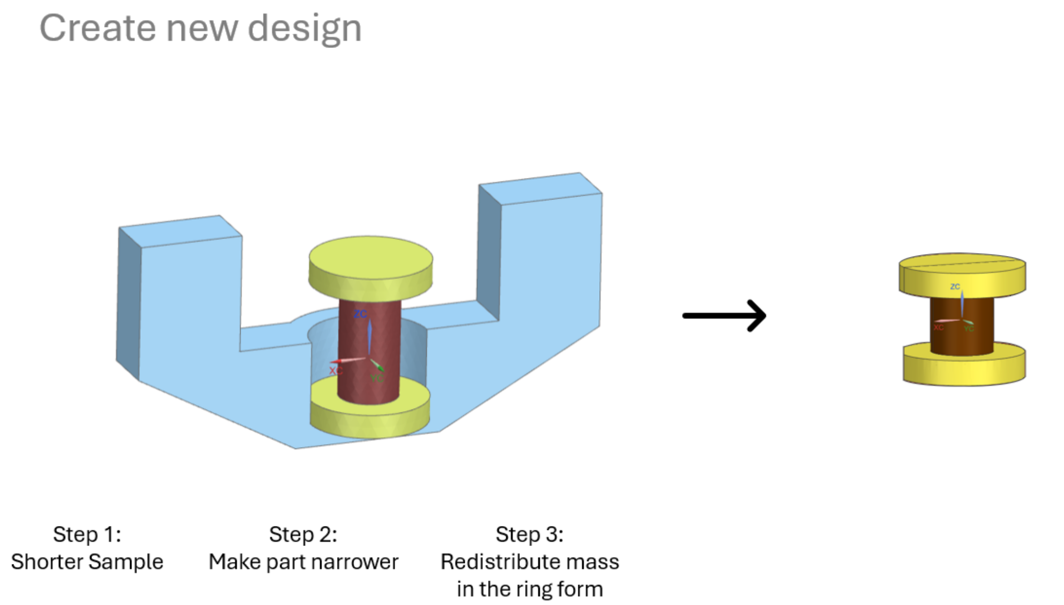
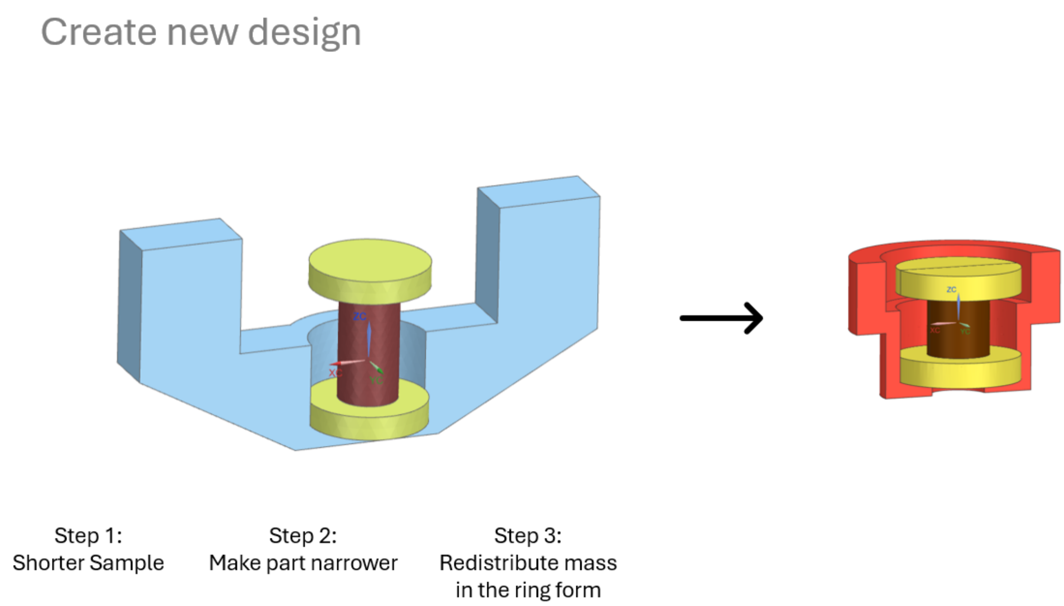
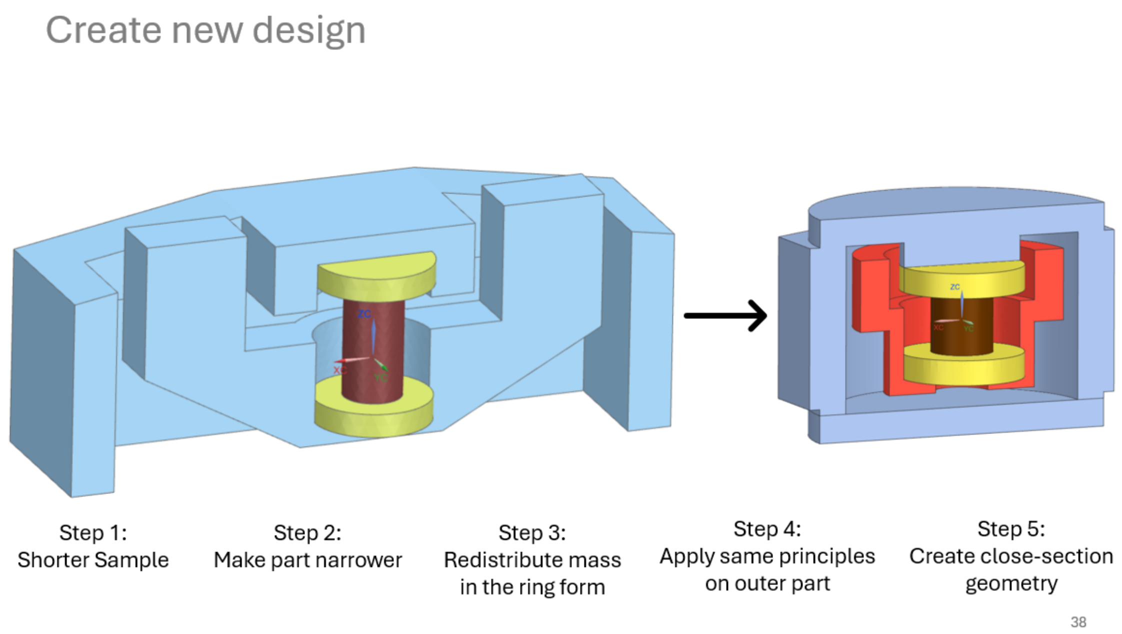
[15:00] So I started the redesign. And I thought: "Okay, uh, what prevents me from making this outer active part, uh, making smaller?" And, uh, it was the inner part because, uh, the... I cannot make it smaller than the component inside. So, okay. What prevents me from making the inner part smaller? And it was the, um, center of gravity alignment. Because we wanted to have our cen... COG in the center of the hose, we had to extend mass on long arms, uh, creating a bulky design. Okay. Yeah? Okay. So, um, I asked myself: "Okay, uh, is there a... can I make the hose shorter? Is there any reason against it?" And the answer was actually yes, I can make it. There was no like really physical validating reason to make it... not to... to keep it this long. It was just a tradition and habit from old test rig generations.



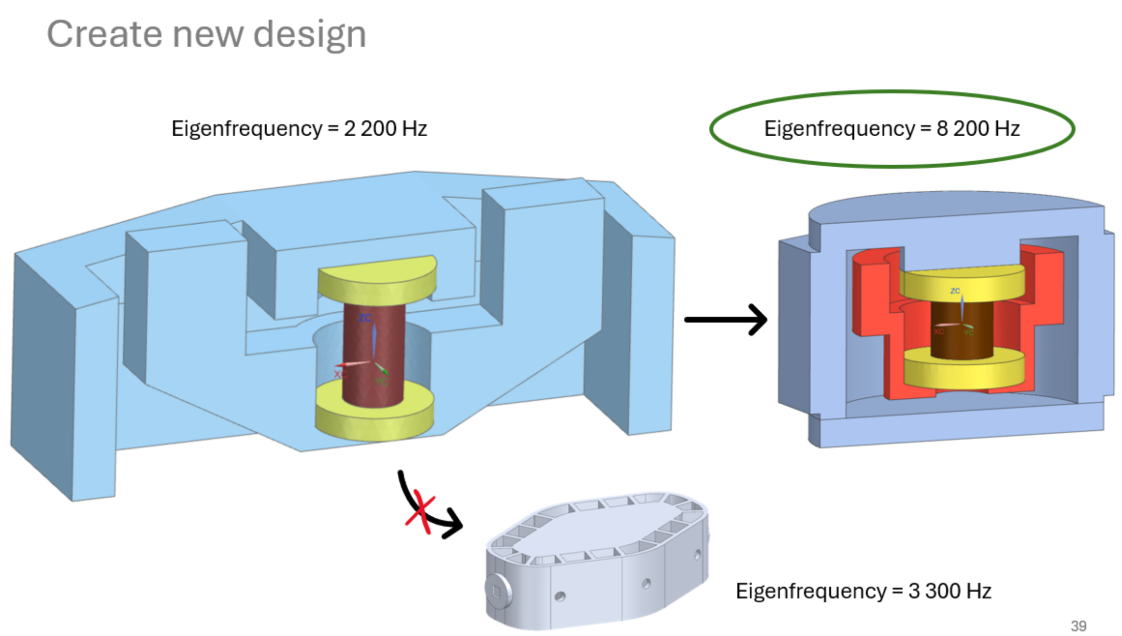
[16:06] So first thing I did, uh, I made the shorter sample which allowed me, um, not to shift the COG that far. Uh, and I didn't stop here. Uh, I tried to... I used this opportunity to completely reimagine the topology. And instead of just shrinking the fork design, I, um, decided to make the part narrower and redistribute the mass in the compact ring form, uh, which resulted in much compactor [sic] geometry and much stiffer geometry.



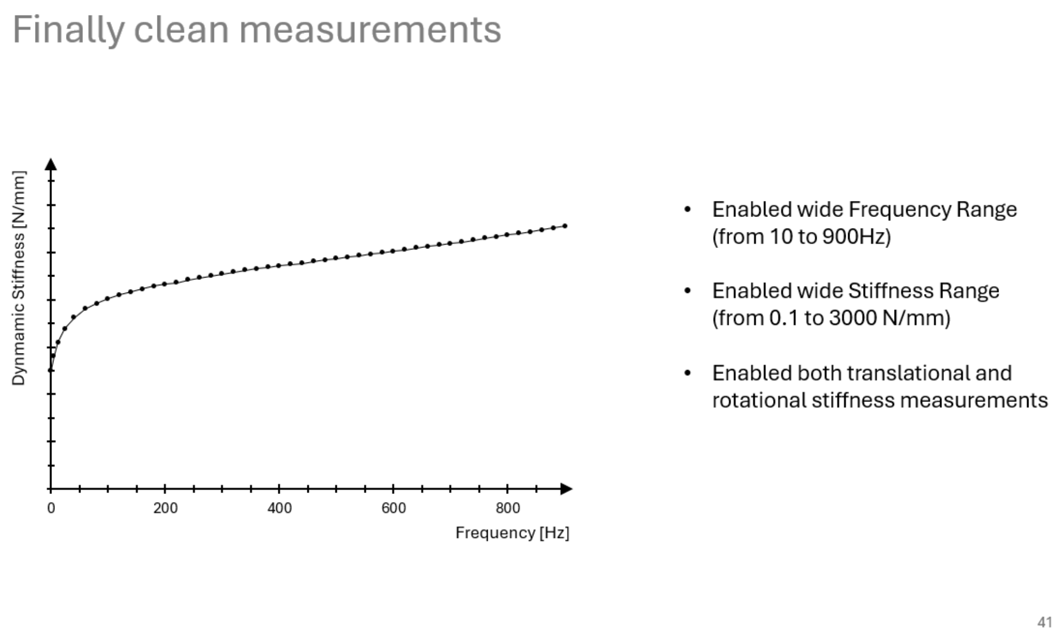
And what is what more importantly even, that I freed a lots of space around it, uh, which allowed me to apply this... the same design approaches to design the active part. Uh, so I managed... I managed to make it much smaller and, uh, achieved this closed geometry.



[16:58] And it resulted a massive boost in the natural frequency. To, uh, take the result into perspective: the, uh, older generation had the frequency 2,200 Hertz. And instead of accepting the, um, incremental improvements to 3,300 Hertz, I managed to jump in one single step to 8,200 Hertz, which is not twice, not three times, but eight times higher than our operational range.



[17:28] And at this point you might ask: "Okay, so what? Did it actually help?" And the answer is: this approach was actually transformative. Uh, for the first time we were able to measure the dynamic stiffness in such a wide frequency range without distortion and with such a precision and smoothness. So, uh, in the end, this innovative measurement method approach developed by my supervisor combined with elegant mechanical design from my side, it, um, resulted a tool which enabled both translational and rotational stiffness measurements, uh, that we can then use for our simulations.



[18:11] And once the tool was ready, I personally conducted over 100 experiments testing everything that came to my mind: uh, thick hoses, thin hoses, rubber mounts, um, plastic hoses, frozen samples, heated samples. And then I built a comprehensive database for the HVAC development team. And what's, um, the very cool in this story I think is that after I left, this tool not just landed somewhere on the shelf, uh, but it became a standard to the team. Uh, and they are still using it today. And I know it because I maintained a close contact, uh, with colleagues while I was working on my Master's thesis at the battery develop… battery development.

Finally clean measurements

- Conducted 100+ experiments
- Characterized diverse materials and geometries across full thermal ranges.
- Build a comprehensive database for the HVAC colleagues.
- Tool became a standard for the team.
- They're still using it today to characterize hoses and compressor mounts.

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[18:57] So basically, um, yes. Um, now I want to quickly summarize, uh, why I succeeded. Not the technical "how," but actually the fundamental "why." Uh, and I think that the answer lies, um, in the... in my engineering courage. Uh, and by engineering courage I mean, um, the courage to reject "we've always done this way." It's the courage to scrap an existing design for a better one. It's a habit of constantly asking "why not" and "what if." Uh, "what if I make the sample shorter?", "what if I make the closed geometry?", "what if I ex... approach the test differently?" And this habit, uh, in combination with digging down to the physical foundations, is actually my personal formula of success in almost every project I take in. And, um, what also inspires me at Tesla very much is that Tesla shares this approach. And I think that it will be a perfect environment for me where my mindset will not face a resistance, but will be appreciated, valued and welcomed.

My Approach to Engineering

- Rejecting "we've always done it this way.“
- Constantly asking "Why not?" and "What if?".
- Digging down to the physical foundations.

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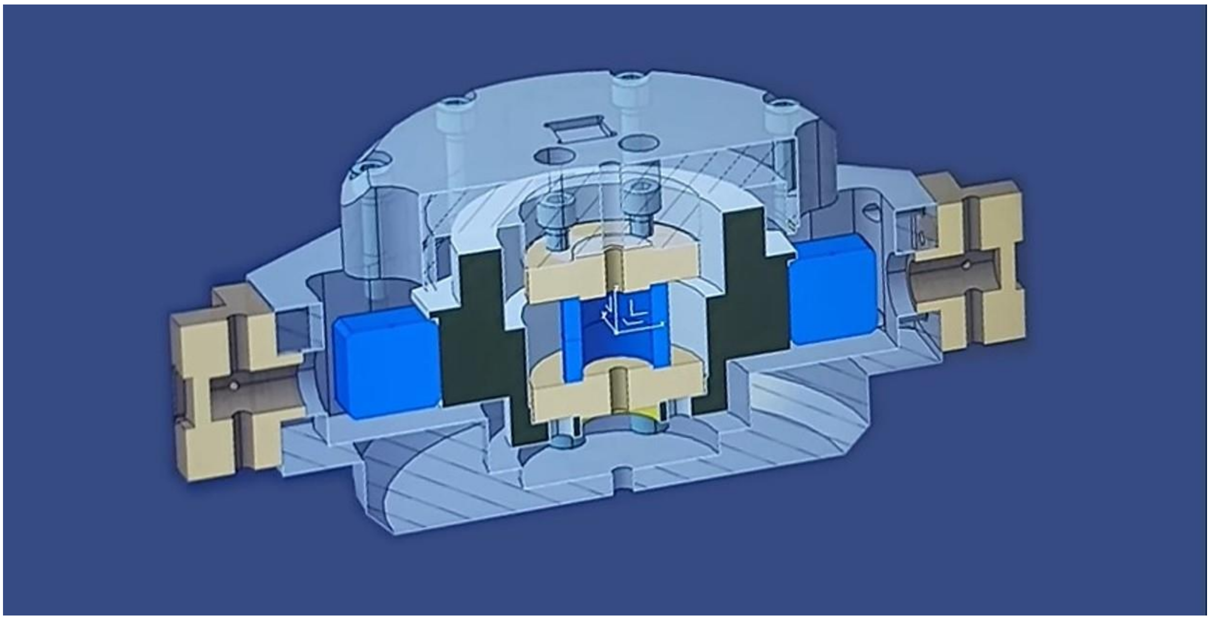
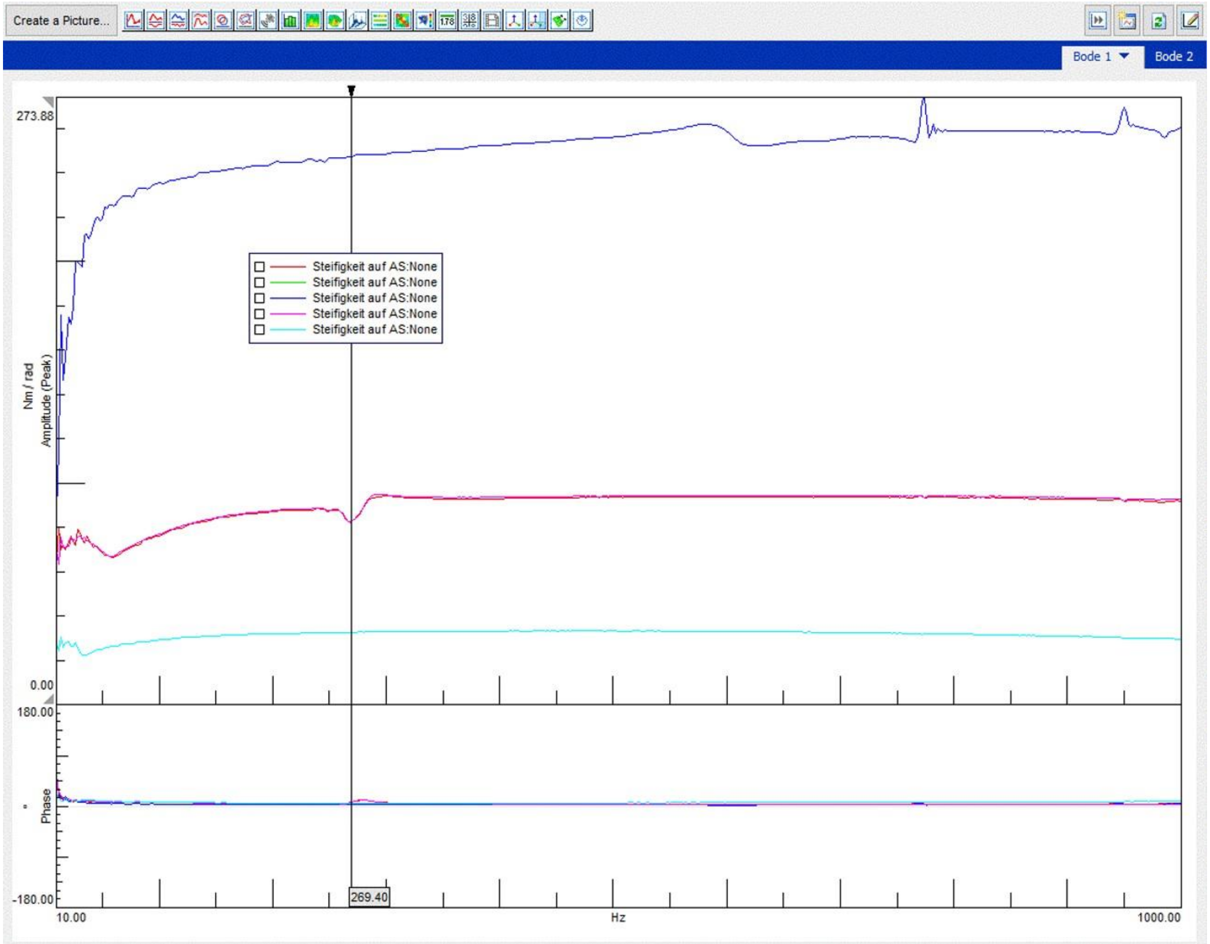
[20:10] So. Yes. That's basically it.

Zusätzliche Fotos:



CAD-Modell des neuen Prüfstands (real, not simplified)

Fotos des gefertigten und aufgebauten Prüfstands im Labor. Shaker, Vibrationssensoren integriert und funktionsbereit.



Bode plot of dynamic torsional stiffness for three refrigerant hose specimens, measured from 10 to 1000 Hz with Simcenter LMS Test.Lab. Dynamic torsional stiffness magnitude on top and loss angle on the bottom — stable, repeatable curves confirm measurement quality.